Leveraging Random Forest Algorithms for Stable Isotope Prediction in Surface Water

Kylie Alfaro[1,2], Stephanie Lyons[1], Olivia Schmitt [1,2], Henrietta Dulai[2], Hailey Pantaleo[2], Casey R McGrath[1]

[1]*Pacific Northwest National Laboratory, Richland, WA 99352, USA*

[2]*Department of Earth Sciences, University of Hawai`i at Mānoa, Honolulu, HI 96822, USA*

Stable water isotope analysis ($\delta^{18}O$ and $\delta^{2}H$) provides insight into local, regional, and global weather patterns, geology, groundwater interactions, and other hydrologic cycle processes. However, long-term monitoring efforts are rare due to sampling challenges, resulting in sparse datasets. Applying machine learning can fill these gaps to enable more robust hydrologic analysis. This project builds and trains a Random Forest to predict $\delta^{18}O$ values from data collected at Mānoa Stream in Honolulu, HI between February 2023 and April 2026, which contains a sparse record of $\delta^{18}O$ and $\delta^{2}H$ measurements alongside other water quality parameters. Random Forest ensembles are well-suited for smaller datasets compared to other algorithms. Leave-One-Out and K-Fold cross validation will quantify predictive accuracy. This approach offers a framework for applying machine learning to sparse environmental datasets across diverse hydrologic settings.